HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed **Python-based data pipelines and ETL scripts** using pandas, SQLAlchemy, and PySpark to automate data extraction and transformation, reducing manual processing time by **45%**.
- Built **RESTful APIs using Flask and FastAPI**, integrating them with AWS services and PostgreSQL, enabling efficient data exchange between internal tools and third-party platforms.
- Designed and maintained backend services in **Java and Spring Boot**, contributing to a 25% improvement in system response time and supporting clean data pipelines for downstream consumers.
- Worked with **PostgreSQL, MySQL, and MongoDB** to optimise queries and improve data retrieval efficiency by 40% for analytics-driven workloads.
- Implemented **automated testing frameworks using pytest** and unittest, achieving 80%+ code coverage and significantly reducing regression defects in Python-based services.
- Deployed Python services on **AWS (EC2, Lambda, S3)** using Docker, and managed batch job scheduling with Jenkins and cron for reliable daily processing.
- Collaborated with data scientists and analysts in an **Agile/Scrum** environment to translate data requirements into robust, production-ready Python pipelines and APIs.
- Wrote clean, modular Python code following **PEP8 standards**, enabling maintainability and easy onboarding of new developers across the team.

SKILLS

Programming Languages – Python (advanced), Java, SQL, Bash scripting.

Python Ecosystem – pandas, NumPy, Flask, FastAPI, SQLAlchemy, PySpark, pytest.

Backend & APIs – Spring Boot, REST API design, microservices, API integration.

Databases – PostgreSQL, MySQL, MongoDB; query optimisation and schema design.

Cloud & DevOps – AWS (EC2, S3, Lambda), Docker, Jenkins, CI/CD pipelines.

Development Practices – Agile/Scrum, Git, TDD, code reviews, clean code principles.

Tools – VS Code, Jupyter Notebook, Postman, PyCharm, IntelliJ IDEA.

PROJECTS

Automated ETL Data Pipeline (HCL Technologies)

- Built a Python-based ETL pipeline using pandas and SQLAlchemy to automate data extraction from multiple sources and load into PostgreSQL, reducing manual effort by 45%.
- Containerised the pipeline with Docker and scheduled automated daily runs via Jenkins, with error alerting and retry logic ensuring reliable and consistent data delivery.

Internal REST API Service (HCL Technologies)

- Developed a FastAPI-based REST service exposing processed internal data to downstream teams; integrated with AWS S3 for file storage and PostgreSQL for persistent data management.
- Implemented request validation, exception handling, and structured logging, reducing API error rates and improving observability for the operations team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)